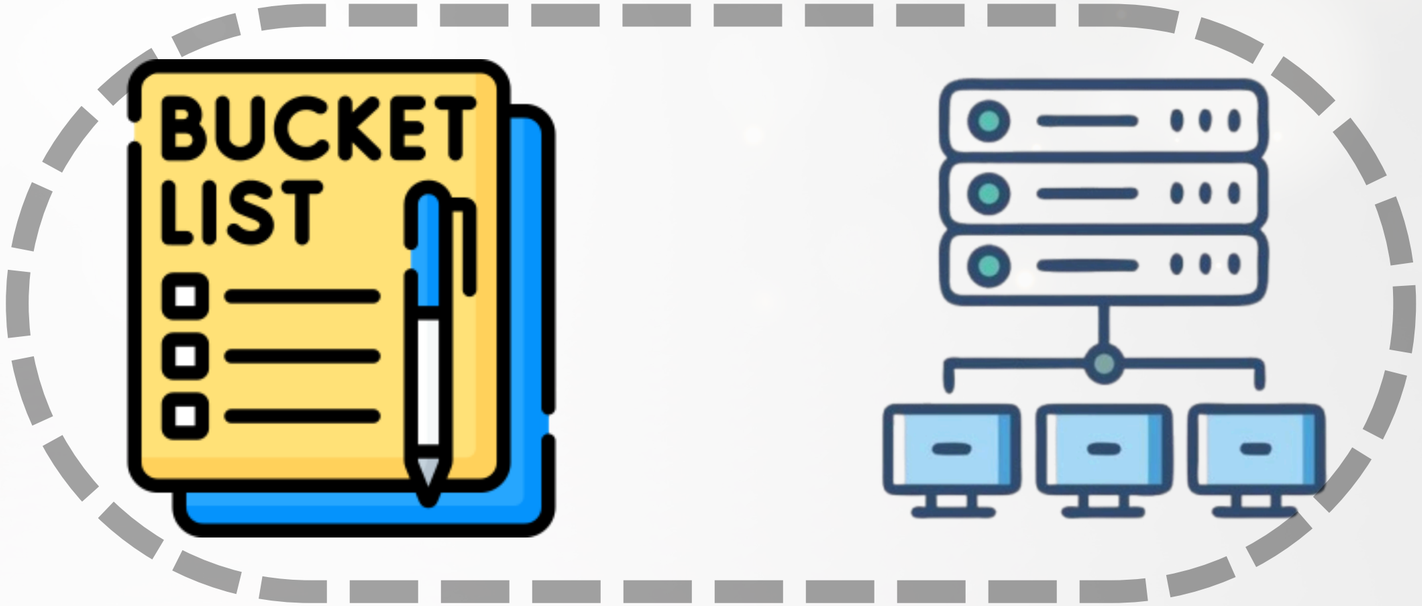
COLLABORATIVE
BUCKET LIST APPLICATION AND TRANSFER PROTOCOL

PROJECT
INTRODUCTION

Project: Collaborative Bucket List Application
Architecture: Client-Server
Transport Layer: TCP
Application-Layer Protocol: CBLP

Main Concept:
A network application that allows multiple users to connect to the same server and collaboratively manage a shared Bucket List.



PROJECT OBJECTIVE

The objective of this project is to develop a network application that allows users to manage a shared Bucket List through a network which allows the user to.

- Create an account system with a username and password
- Properly authenticate before using the application
- Add/Remove/Edit shared Bucket List items and description
- View the shared Bucket List among users
- See other users currently connected to the system
- Allow multiple users to use such functions in shared bucket list at the same time while also notifying every other clients of the user's activity

APPLICATION STRUCTURE

1

CLIENT

- CONNECTS TO SERVER USING TCP
- ACCEPTS USER COMMANDS
- CREATES CBLP REQUESTS
- DISPLAY RESPONSE AND NOTIFICATIONS

2

SERVER

- ACCEPTS TCP CONNECTIONS
- HANDLES MULTIPLE CLIENTS
- PROCESSES CBLP REQUESTS
- MAINTAINS ACCOUNTS AND BUCKET LIST
- SENDS RESPONSES AND NOTIFICATIONS

3

PROTOCOL

- DEFINES CBLP MESSAGE STRUCTURE
- DEFINES STATUS CODES AND PHRASES
- ENCODES/DECODES JSON MESSAGES

APPLICATION CHARACTERISTIC

CLIENT-SERVER ARCHITECTURE

A central Server manages users and the shared Bucket List.

MUTLIPLE-CLIENTS SUPPORT

Multiple users can connect to the Server simultaneously.

SHARED APPLICATION STATE

The Bucket List is maintained by the Server and shared between clients.

USER AUTHENTICATION SYSTEM

Users must register and authenticate before using the Bucket List.

REUQEST-RESPONSE COMMUNICATION

Clients send requests and the Server returns responses.
The Server can also notify other connected clients when the shared list changes.

SERVER AND CLIENT ROLES IN APPLICATION

SERVER

- Managing accounts
- Authenticating users
- Maintaining the Bucket List
- Processing requests
- Sending responses
- Broadcasting changes to connected clients

CLIENT

- Accepting commands list from the server
- Sending requests to the server
- Receiving and displaying responses
- Displaying noficiations

TRANSPORT LAYER: TCP

This project uses TCP (Transmission Control Protocol) as the main Transport Layer service

Why TCP?
The application contains various information that needs to be delivered correctly such as.

- Username
- Password
- Bucket List items
- Descriptions
- Add/Edit/Remove operations

Losing or corrupting this information could cause the shared application state to become **incorrect**.
Therefore, **reliable communication** is **much** more important than minimizing latency.

TCP ADVANTAGES OVER UDP

1

RELIABLE DATA TRANSFER

TCP PROVIDES RELIABLE DELIVERY OF DATA.

2

ORDERED DELIVERY

MESSAGES ARRIVE AT APPLICATION IN THE SAME ORDER.

3

ERROR HANDLING

TCP HANDLES PACKET LOSS AND RETRANSMISSION AUTOMATICALLY.

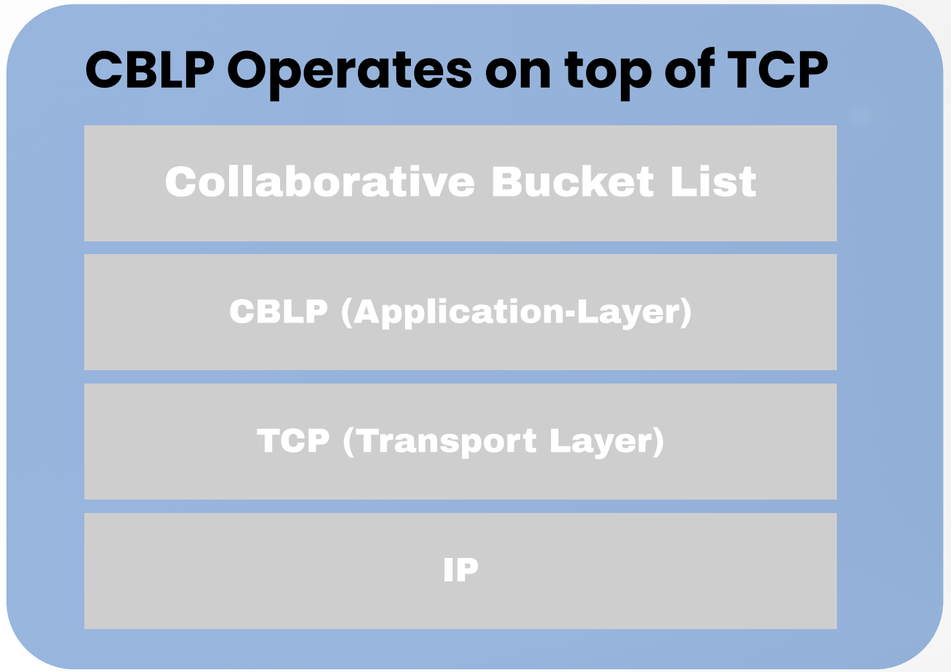
APPLICATION LAYER PROTOCOL: CBLP

Collaborative Bucket List Protocol

CBLP is this project's Application-Layer Protocol designed specifically for the Collaborative Bucket List Application.

CBLP Defines

- Request types
- Request format
- Response format
- Status codes
- Status phrase
- Payload structure
- Server notification



CBLP MESSAGE FORMATS

Collaborative Bucket List Protocol uses “JSON” as it‘s message format.

Request (Client › Server)

```
{
    "request": "ADD",
    "payload": {
        "title": "Visit Japan",
        "description": "Travel around Tokyo and Kyoto"
    }
}
```

Response (Server › Client)

```
{
    "status": 201,
    "phrase": "CREATED",
    "request": "ADD",
    "payload": {}
}
```

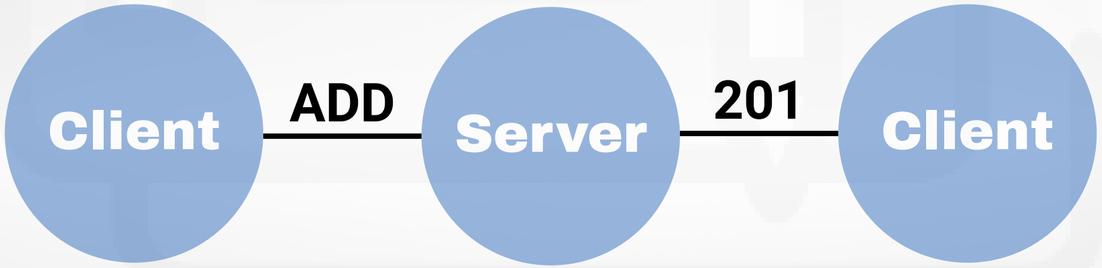
```
[201 CREATED] ADD
{
  "item": {
    "id": 1,
    "title": "Visit Japan",
    "description": "Travel around Tokyo and Kyoto",
    "added_by": "Alice"
  }
}
```

CBLP AVAILABLE REQUESTS

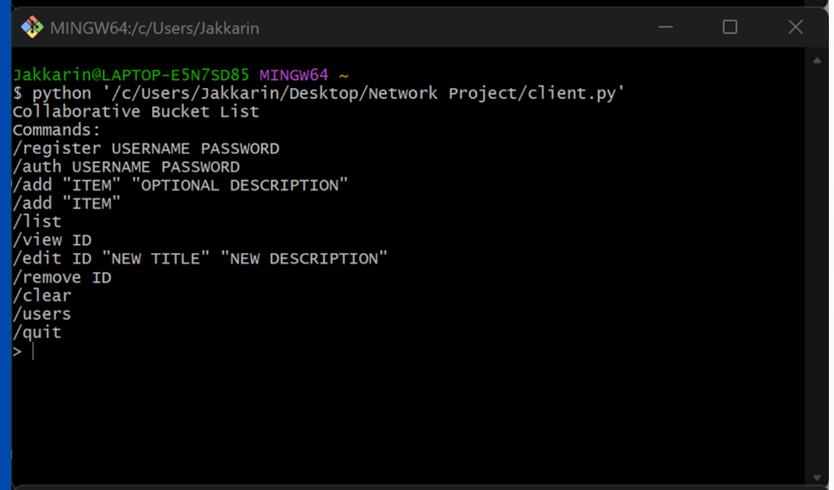
REGISTER	CREATE AN ACCOUNT
AUTH	AUTHENTICATE A USER
ADD	ADD BUCKET LIST ITEM
LIST	DISPLAY ALL ITEMS
VIEW	DISPLAY ONE ITEM
EDIT	EDIT AN ITEM
REMOVE	REMOVE AN ITEM
CLEAR	CLEAR THE LIST
USERS	DISPLAY CONNECTED USERS
QUIT	END CONNECTION

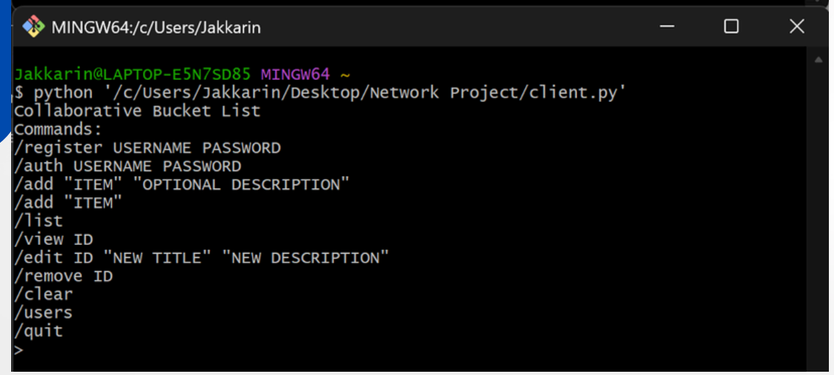
RESPONSE STATUS CODES

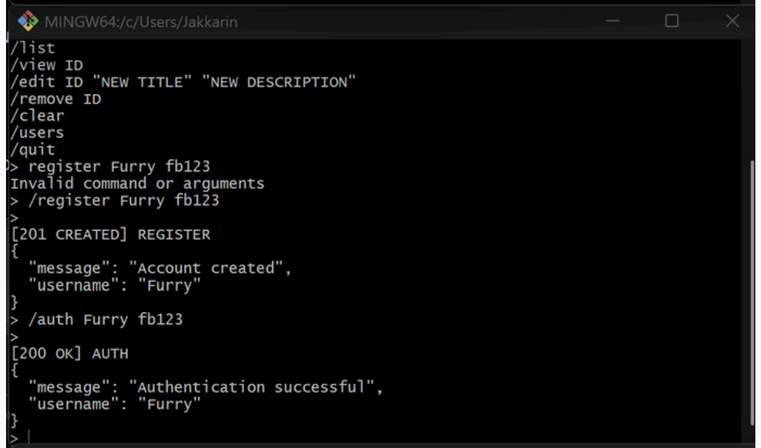
200	OK	REQUEST SUCCESSFUL
201	CREATED	NEW RESOURCE CREATED
400	BAD_REQUEST	INVALID REQUEST/DATA
401	UNAUTHORIZED	AUTHENTICATION FAILED/REQUIRED
404	NOT_FOUND	RESOURCE DOES NOT EXIST
409	CONFLICT	REQUEST CONFLICT WITH STATE
500	INTERNAL_SERVER_ERROR	UNEXPECTED SERVER ERROR

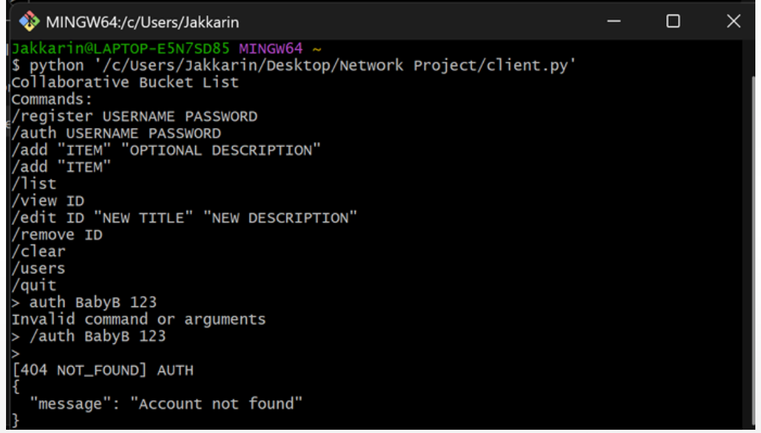


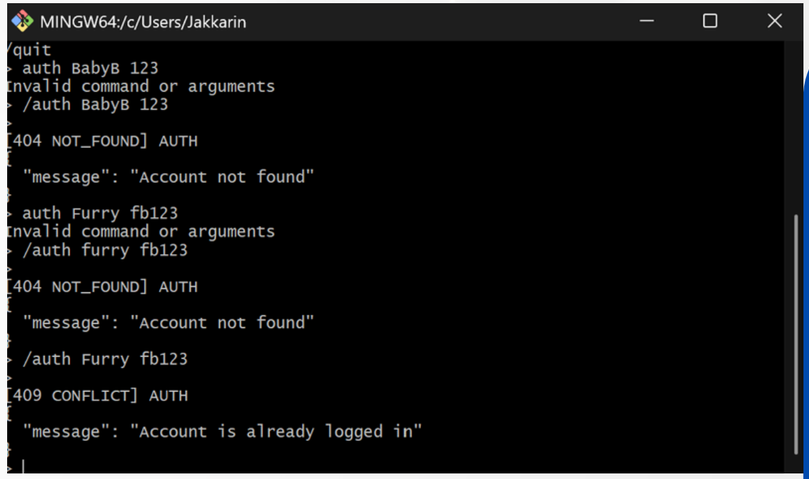
USAGE DEMONSTRATION

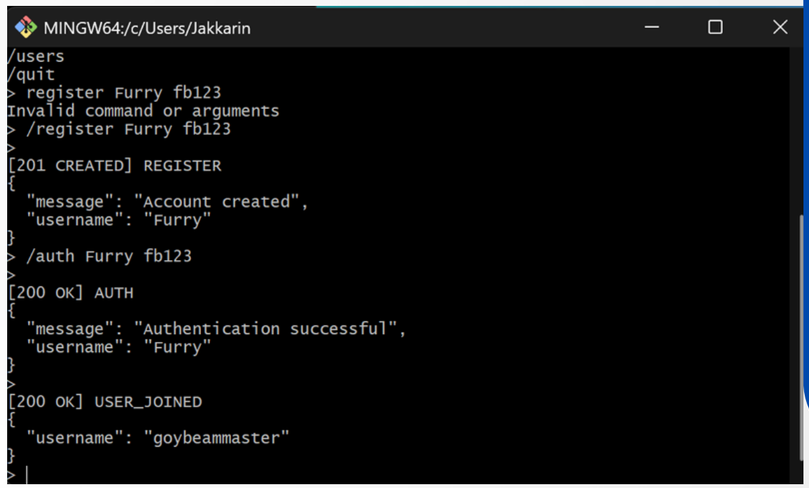






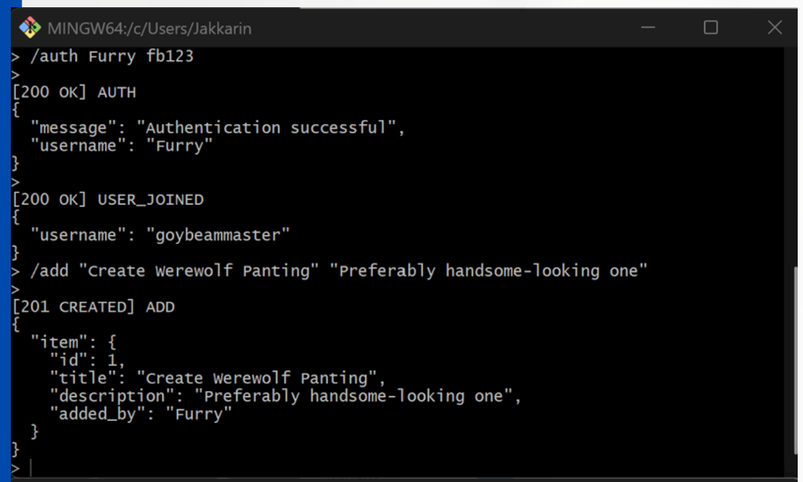


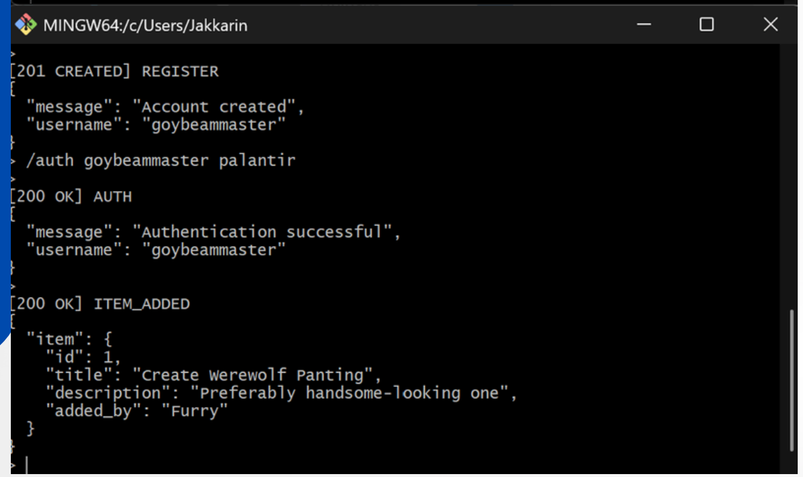




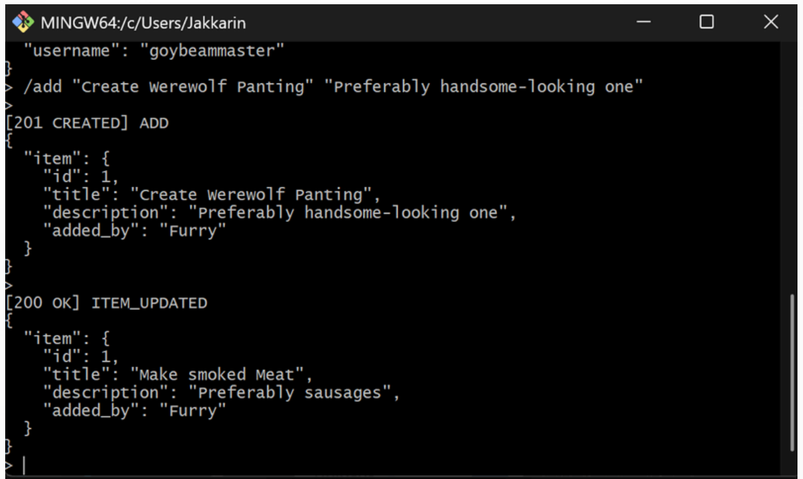
USAGE DEMONSTRATION

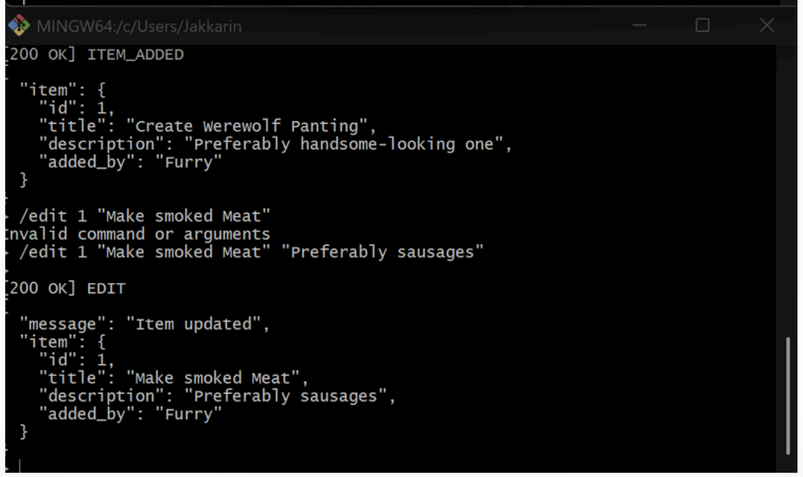
ADD REQUEST



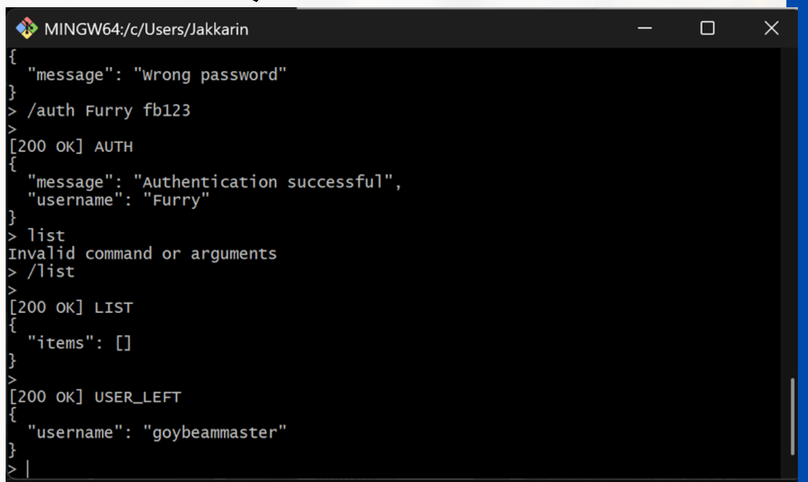


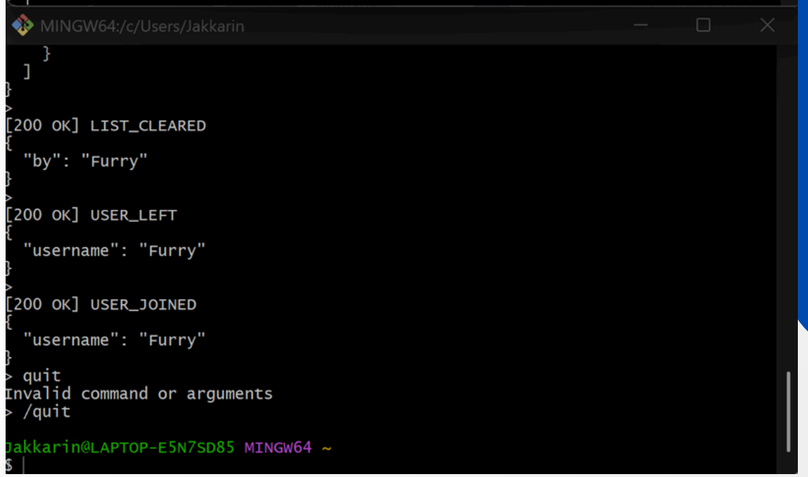
EDIT REQUEST





LIST,QUIT REQUEST





THANK YOU